

Planet Wars Competition Entry

Tohoku **AI** Lab

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Agent Overview

- Core concept: “Standard” PPO RL agent
- Reward: Planet ownership
- Key Novelty: Inductive bias through semantic actions
 - Which decision are you more likely to make?
 - Send 14.7% of ships from A to B
 - Send enough ships from A to capture B
 - Actions: capture/defend/send all/send surplus
 - See Felix Neumann’s 3rd place Orbit Wars writeup on Kaggle

System Design

- 100% vibe coded
- Engine re-written in Jax – end-to-end GPU training
- Strategy: Pure self-pay (PFSP)
 - Checkpoint pool + active roster
 - Periodically re-evaluate entire pool
 - FIFO eviction

```
network: 55,139 parameters (0.055M)
  MultiHeadDotProductAttention_0    16,640
  Dense_2                            8,320
  Dense_8                            8,320
  Dense_3                            8,256
  Dense_10                           4,160
  Dense_0                            2,240
  Dense_6                            2,080
  Dense_7                            2,080
  Dense_9                            2,080
  Dense_1                             512
  LayerNorm_0                        128
  LayerNorm_1                        128
  Dense_4                             65
  Dense_5                             65
  Dense_11                            65
```

Results

- Agents
 - tail-test – non-PFSP **trained against scripted agents** for evaluation
 - tail-warm-start – non-PFSP **trained against tail-test** for evaluation
 - tail-small – PFSP only
 - ~3.5B environment steps in 1 – 2 days
- SP has 100% win rate against all scripted agents in < 10 mins

Analysis and Insights

- Sparse or ship-only rewards didn't work well
- Non-semantic actions led to extremely slow training
- PFSP implementation needs tuning
 - Sometimes plateaus
 - A start/stop of training (due to shared GPU) usually provided a boost in performance
- Typical LLM problems, but still an extremely valuable tool

Future Work

- Limitations
 - FIFO isn't a great pool eviction policy
 - No time to look into hyperparameters (PPO, network)